Computational Physics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University
setu.ie

Computational Physics (A13987)

Short Title:	Computational Physics		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	KMURPHY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The aim of this module is to highlight the computational approach to solving physics problems. The concepts and techniques covered in this module will be applicable in projects or problem-solving work that contain a strong computational or data analysis element. The module is designed such that a significant fraction of the student's time is spent programming/solving specific physical problems, rather than learning abstract techniques.

Programmes

`PHYI-0038`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	4 / 7 / M

Indicative Content

- Introduction to scientific programming: symbolic versus numeric computation, floating point numbers and error analysis, machine precision, general classification of computational problems
- Overview of computational problems: integration and differentiation, interpolation and extrapolation, ordinary differential equations, partial differential equations, matrix methods, curve fitting, Monte Carlo other simulation methods
- Scientific programming tools: Python (or an equivalent language, such as Julia, Java or MATLAB/ Octave), LaTeX, Markdown, etc.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discriminate between and apply essential methods and techniques (error analysis, problem classification) for symbolic/numerical computation in physics.
2. Design and implement simulation methods, such as Monte Carlo, to solve deterministic as well as probabilistic physical problems.
3. Evaluate given computational physics problems and design a suitable implementation with appropriate error analysis.
4. Select and use appropriate computational software tools to obtain/collate scientific data, perform analysis and generate reports.

Learning and Teaching Methods

- The module is covered through a series of practical problem solving sessions and directed independent learning.
- Due to the very practical nature of the skills to be acquired in this module, these practical sessions will be centred around the idea of learning by doing, whereby students develop proficiency in the specified skill set through guided activities, and whereby lecturers provide short formal presentations of relevant concepts and technologies, as well as practical tips, feedback, and advice on best practices.
- Active engagement with frequent practice on examples is strongly encouraged through regular course work consisting of a mixture of practical activities graded from guided to self-directed tasks.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
Assignment	30%	1,4
Assignment	30%	1,3,4
Assignment	40%	1,2,3,4

Assessment Criteria

<40%: Demonstrate little or no evidence of understanding of the concepts outlined in the syllabus content and the skills required for attaining the module learning outcomes. Show very little or no ability to apply knowledge to solve problems. Organisation and presentational skills are minimally effective or ineffective.

40%-49%: Demonstrate partial but limited command of knowledge and skills required for attaining some of the module learning outcomes. Able to implement a rudimentary solution to the given computational problems.

50%-59%: Demonstrate general but incomplete command of knowledge and skills required for attaining most of the module learning outcomes. Shows evidence of some analytical and critical abilities, and ability to apply knowledge to most familiar computational problems.

60%-69%: Demonstrate substantial command of a broad range of knowledge and skills required for attaining at least most of the module learning outcomes. Show evidence of analytical and critical abilities, and ability to apply knowledge to familiar and some unfamiliar computational problems. Correct use of data/results to draw appropriate conclusions.

70%-100%: Demonstrate command at an advanced level of the knowledge and skills required for attaining all the course learning outcomes. Show strong analytical and critical abilities, with evidence of original thought, and ability to apply knowledge to a wide range of complex, familiar and unfamiliar computational problems. Critical use of data and results to draw appropriate and insightful conclusions.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	36	
Independent Learning	99	

Supplementary Material

- Landau, R. _Computational Physics: Problem Solving with Python_. NY: Wiley, 2015.
- Newman, M. _Computational Physics with Python_. USA: CreateSpace Independent Publishing Platform, 2012.

Requested Resources

- Room Type: Computer Lab